

Constrained Geo-Semantic Identifiers and Utility-Supervised Dynamic Multi-Expert Retrieval for Next-POI Prediction

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Abstract—Next point-of-interest (POI) prediction must select a destination from a large, heterogeneous catalog in which many locations are only sparsely observed. We present GeoSemID, a standalone retrieval-enhanced framework organized around three contributions. First, a hierarchical geo-semantic identifier augments each atomic POI with constrained geographic, categorical, and semantic codes, giving long-tail locations reusable structure for statistical backoff. Second, a utility-supervised dynamic multi-expert retrieval mechanism couples four complementary experts—spatial, transition, temporal, and semantic—through a sample-level router whose supervision is derived from training-target ranks, and fuses their evidence on a common rank-percentile scale. Third, an evidence-preserving, candidate-constrained ranking stage retains expert evidence and scores only valid candidate identifiers with a language model, avoiding open-ended POI-ID generation. Across 1,818 CA, 988 NYC, and 2,206 TKY test instances, GeoSemID reaches 42.48%, 44.65%, and 46.85% HR@10, improving over the strongest competing baseline by 3.58, 3.80, and 4.25 percentage points, respectively. Its dynamically fused top-50 candidate pool attains 60.00%, 62.40%, and 64.20% target recall on the three datasets.

Index Terms—Next point-of-interest prediction, trajectory modeling, semantic identifiers, multi-expert retrieval, dynamic routing, large language models.

I. INTRODUCTION

Next point-of-interest (POI) prediction is a fundamental problem for intelligent transportation systems, location-based services, and personalized mobility assistance. A useful predictor must reconcile persistent user preferences with immediate movement intent while accounting for temporal regularity [1], geographic reachability, and the functional meaning of locations. Although sequential and spatiotemporal models [2]–[4] have substantially improved trajectory encoding, the decision space remains a large catalog of heterogeneous POIs.

Atomic POI identities provide exact discrimination but expose little reusable structure for sparse locations. A long-tail POI may be geographically close to, functionally similar to, or used in a context comparable with better-observed locations, yet these relations are not automatically available to an identifier-only model. Metadata and continuous embeddings provide important signals and are strong baselines; however, they do not by themselves define discrete, reusable levels for statistical backoff across geographic, categorical, and contextual relations. In the benchmarks we study, a large majority of POIs—about 70%—are visited only a handful of times, so transferring knowledge to these long-tail locations is decisive for end-to-end accuracy.

Candidate generation is therefore not merely a preprocessing detail. Spatial, transition, temporal, and semantic signals capture different mobility constraints, and the useful signal varies across trajectory instances. A single retriever or fixed fusion rule can miss the target; once the target is absent from the candidate set, a downstream ranker cannot recover it. This issue is particularly consequential when a language model is used to reason over heterogeneous trajectory context but must still choose from a large, discrete POI catalog.

We address these issues with GeoSemID, a retrieval-enhanced framework that couples structured POI representation with sample-level candidate generation. GeoSemID augments the atomic POI identity with constrained geographic, categorical, and semantic codes. Spatial, transition, temporal, and semantic experts retrieve complementary evidence; a utility-supervised router learns their instance-specific reliability, and rank-percentile fusion makes their outputs comparable. The resulting expert evidence is retained by a candidate-constrained LLM ranker rather than discarded after retrieval.

These capabilities are directly relevant to intelligent transportation systems. Accurate next-POI prediction feeds travel-demand estimation and traffic-flow forecasting, powers proactive location-based services such as context-aware routing and personalized recommendation, and supports mobility-as-a-service planning, where anticipating a traveler’s next stop improves dispatching and resource allocation. Robustly handling sparse, long-tail destinations is essential in these settings, because operational value often concentrates on infrequent but high-impact trips.

The contributions of this paper are threefold:

- 1) **Hierarchical GeoSemID construction.** We introduce GeoSemID, a constrained geo-semantic identifier that preserves atomic POI identity while supplying reusable coarse/fine geographic, categorical, and semantic codes, enabling structured statistical backoff for sparse and long-tail locations.
- 2) **Utility-supervised dynamic multi-expert retrieval.** We couple four complementary experts (spatial, transition, temporal, and semantic) with a sample-level router supervised by training-target ranks, and combine their outputs through evidence-preserving dynamic fusion on a common rank-percentile scale, replacing fixed or globally tuned combination rules.
- 3) **Evidence-preserving candidate-constrained ranking.** We develop a ranking protocol that carries expert evidence forward and scores only valid candidate identifiers with

a language model, separating candidate coverage from conditional discrimination and avoiding open-ended POI-ID generation.

II. RELATED WORK

A. Spatiotemporal Modeling and Semantic POI Representation

Early sequential recommendation models such as factorized personalized Markov chains model the dependence between prior and subsequent choices [5]. For mobility, spatial and temporal contexts were explicitly incorporated into recurrent prediction by ST-RNN [6], while attention-based trajectory encoders such as DeepMove [7] improved the use of historical mobility patterns. More recent attention and flow-aware models, including STAN [8], LSTPM [9], and GETNext [10], further model nonlocal spatiotemporal dependencies. These studies substantially improve how trajectories are encoded, but a stronger trajectory encoder does not by itself determine how geographic, functional, and semantic relations among candidate POIs are represented and shared.

Many representative methods primarily rely on atomic POI identities, category features, coordinates, or shallow combinations of these signals. Continuous semantic embeddings remain a valuable way to express similarity, whereas discrete identifiers can additionally provide reusable symbolic structure. GeoSemID complements rather than replaces atomic identity: it makes geographic, categorical, and constrained semantic codes available to retrieval, backoff, and ranking components. This positioning does not assume that existing embeddings are weak; it targets the different need for explicit, shareable levels under sparse mobility statistics.

B. Candidate Retrieval and LLM-based Ranking for POI Recommendation

Large POI catalogs motivate retrieve-then-rank pipelines because candidate generation determines the feasible decision set of a later ranker. Geographic neighborhoods, transition statistics, temporal preference, and vector similarity provide complementary retrieval signals, yet a set union or a globally fixed weighting cannot express that a routine commute and an exploratory visit may require different evidence. Prior LLM-based POI prediction pipelines explore staged designs that couple trajectory understanding with candidate refinement and language-model prediction [11]–[13], yet they typically rely on fixed retrieval sources and hand-designed coordination rather than learned, sample-level fusion.

LLM-based POI systems can serialize long-term profiles, recent trajectories, time, and location metadata into a common interface, but they still face a large discrete output space and candidate-order/context-length effects. Candidate-constrained scoring offers a more controllable alternative to unconstrained catalog generation. Retrieval provenance is also not always explicitly retained by the downstream ranking interface. GeoSemID therefore learns expert utility at the trajectory-instance level, fuses expert rankings on a common scale, and passes each retained candidate’s expert evidence to

the final constrained ranker. This design treats retrieval reliability and ranking discrimination as coupled stages without making RRF part of the proposed inference method.

C. POI Representation, Expert Fusion, and Constrained Generation

Beyond trajectory encoding, a complementary line of work studies how to *represent* POIs so that geographic and functional relations become reusable. Geographic knowledge graphs and region-embedding methods [14]–[17] inject spatial adjacency and area semantics into location representations [18], while discrete tokenization and semantic identifiers offer a symbolic alternative to purely continuous embeddings [19]–[22]. Continuous embeddings capture graded similarity, whereas discrete codes expose explicit levels for statistical backoff [23]; GeoSemID combines both by attaching constrained hierarchical codes to each atomic POI identity.

Mixture-of-Experts architectures and learned gating [24]–[26] route inputs to specialized sub-models, with sparse and multi-gate variants scaling this idea to large and multi-task models [27]–[29]. Rank-fusion techniques such as reciprocal-rank fusion aggregate heterogeneous ranked lists into a single ordering [30], [31], while learning-to-rank methods optimize such orderings directly from supervision [32]–[34]. Most POI retrieval systems, however, apply fixed or globally tuned fusion weights that cannot adapt to per-trajectory reliability. GeoSemID instead supervises a sample-level router with training-target ranks, turning static fusion into a utility-aware dynamic combination.

Two-stage retrieve-and-rank pipelines are standard in large-scale recommendation and information retrieval, where an efficient retriever narrows the decision space before an expressive ranker reorders candidates [35]–[38]. When the ranker is a language model, constrained decoding and candidate-scoring techniques keep generation inside a valid output space and avoid hallucinated identifiers [39]–[42]. GeoSemID adopts this discipline by scoring only valid candidate identifiers and, distinctively, by preserving each candidate’s expert evidence for the ranker.

These directions have largely advanced in isolation. GeoSemID unifies structured geo-semantic representation, utility-supervised dynamic multi-expert fusion, and evidence-preserving constrained ranking within a single standalone framework. This integrated design supersedes the fixed-fusion and representation-agnostic assumptions of prior pipelines.

III. PROBLEM FORMULATION

A. Next-POI Prediction

Let $\mathcal{U} = \{u_1, \dots, u_{|\mathcal{U}|}\}$ and $\mathcal{P} = \{p_1, \dots, p_{|\mathcal{P}|}\}$ denote the user and POI sets. Each POI p is associated with metadata

$$\mathbf{m}_p = \langle n_p, c_p, \mathbf{g}_p, \mathbf{a}_p \rangle, \quad (1)$$

where n_p is the POI name, c_p is its functional category, $\mathbf{g}_p$ contains coordinates and available administrative information, and $\mathbf{a}_p$ denotes other attributes.

A check-in of user u at sequence position i is

$$x_{u,i} = \langle p_{u,i}, t_{u,i} \rangle. \quad (2)$$

For a sample whose target is $p_u^* = p_{u,n_u}$, the observable trajectory is $\{x_{u,1}, \dots, x_{u,n_u-1}\}$. Given a recent-window length L , it is divided into

$$\mathcal{T}_u^H = \{x_{u,1}, \dots, x_{u,n_u-L-1}\}, \quad (3)$$

$$\mathcal{T}_u^C = \{x_{u,n_u-L}, \dots, x_{u,n_u-1}\}, \quad (4)$$

where $\mathcal{T}_u^H$ characterizes long-term mobility preference and $\mathcal{T}_u^C$ describes the immediate movement context.

Because the complete POI catalog contains many irrelevant alternatives, we formulate next-POI prediction as two coupled tasks. First, a retrieval model constructs a compact candidate set $\mathcal{C}_u \subset \mathcal{P}$ and should maximize

$$\Pr(p_u^* \in \mathcal{C}_u \mid \mathcal{T}_u^H, \mathcal{T}_u^C, \{\mathbf{m}_p\}_{p \in \mathcal{P}}). \quad (5)$$

Second, a ranker predicts

$$\hat{p}_u = \arg \max_{p \in \mathcal{C}_u} P_\psi(p \mid \mathcal{T}_u^H, \mathcal{T}_u^C, \mathcal{C}_u, \text{GeoSemID}(p), \xi_{u,p}), \quad (6)$$

where $\text{GeoSemID}(p)$ is the structured identifier of candidate p and $\xi_{u,p}$ records expert retrieval evidence. If $p_u^* \notin \mathcal{C}_u$, the end-to-end prediction is necessarily incorrect; therefore candidate coverage and conditional ranking quality must be evaluated separately.

B. Constrained Geo-Semantic Identification

An atomic POI ID preserves exact identity but does not expose reusable geographic, functional, or semantic relations. For each POI, we define

$$\text{GeoSemID}(p) = [z_p^{\text{cg}}, z_p^{\text{fg}}, z_p^{\text{cat}}, z_p^{\text{sem}}], \quad (7)$$

where z_p^{cg} and z_p^{fg} denote coarse- and fine-grained geographic cells, z_p^{cat} is a normalized functional code, and z_p^{sem} is a constrained semantic code.

The first three components are deterministic functions of metadata:

$$z_p^{\text{cg}} = \rho_{\text{cg}}(\mathbf{g}_p), \quad z_p^{\text{fg}} = \rho_{\text{fg}}(\mathbf{g}_p), \quad z_p^{\text{cat}} = \rho_{\text{cat}}(c_p). \quad (8)$$

The semantic component is produced offline by a fine-tuned LLM generator:

$$z_p^{\text{sem}} = \text{Validate}[G_\phi(\mathcal{I}(\mathbf{m}_p))], \quad (9)$$

where $\mathcal{I}(\cdot)$ serializes POI metadata under a fixed prompt, G_ϕ is the generator, and $\text{Validate}(\cdot)$ enforces the predefined grammar and vocabulary. The generator performs a single constrained metadata-to-code mapping; it is never invoked as an interactive or self-directed module and produces one code per POI in a single offline pass.

The semantic-code target has the fixed form

$$y_p^{\text{sem}} = \langle y_p^{\text{intent}}, y_p^{\text{attribute}}, y_p^{\text{context}} \rangle \in \mathcal{V}_i \times \mathcal{V}_a \times \mathcal{V}_c. \quad (10)$$

The category field is deliberately excluded from this code: it is represented only by z_p^{cat} . The semantic fields instead encode visit intent, distinctive service attributes, and contextual usage beyond the source taxonomy. The label-construction protocol, vocabulary sizes, and quality-control statistics are reported with the experimental settings.

C. Context-Adaptive Multi-Expert Retrieval

Let

$$\mathcal{M} = \{\text{sp}, \text{tr}, \text{tp}, \text{se}\} \quad (11)$$

denote spatial, transition, temporal, and semantic experts. Each expert scores the complete catalog during offline training and returns its top- K_0 list during inference:

$$\mathcal{C}_u^{(m)} = \text{TopK}_{K_0}\{r_u^{(m)}(p) : p \in \mathcal{P}\}. \quad (12)$$

A sample-level router estimates $\pi_u = \{\pi_u^{(m)}\}_{m \in \mathcal{M}}$ from observable trajectory context only. The router fuses rank-normalized expert evidence into the final $\mathcal{C}_u$. Training-target ranks are used only to construct router supervision on the training split and are never inputs to the router at validation or test time.

IV. METHODOLOGY

A. Framework Overview

The framework contains three coupled stages. First, GeoSemID converts heterogeneous POI metadata into a reusable hierarchy while preserving the original POI ID for exact identity. Second, transition and temporal retrievers use count-reliability-aware hierarchical backoff, whereas spatial and semantic retrievers use calibrated hierarchical evidence; a target-rank-supervised router dynamically selects their relative importance for each sample. Third, a candidate-constrained LLM scores only valid candidate identifiers and retains expert evidence in its input.

The novelty boundary is therefore not “using Qwen” or merely combining four retrievers. The intended technical contributions are: (i) a constrained, reusable geo-semantic code that enables structured backoff for sparse POIs; (ii) sample-level utility supervision for the dynamic router rather than fixed fusion; and (iii) evidence preservation from retrieval to candidate ranking.

B. Offline GeoSemID Construction

1) *Semantic-Code Supervision and Data Isolation*: The semantic generator is trained on

$$\mathcal{D}_{\text{sem}} = \{(\mathcal{I}(\mathbf{m}_p), y_p^{\text{sem}})\}, \quad (13)$$

using token-level negative log-likelihood

$$\mathcal{L}_{\text{sem}} = - \sum_{(\mathbf{x}, \mathbf{y}) \in \mathcal{D}_{\text{sem}}} \sum_{\ell=1}^{|\mathbf{y}|} \log P_\phi(y_\ell \mid \mathbf{x}, y_{<\ell}). \quad (14)$$

The target serialization is fixed, for example,

$$\langle \text{INTENT}=\text{i}; \text{ATTRIBUTE}=\text{a}; \text{CONTEXT}=\text{c} \rangle, \quad (15)$$

and decoding accepts only tokens in the corresponding field vocabularies. An output is valid if and only if it contains exactly one value for every field and every value belongs to the predefined vocabulary. Invalid outputs are regenerated once under greedy decoding; a second invalid output is mapped to an explicit UNK code and counted in the reported invalid-output rate.

The three field vocabularies $\mathcal{V}_i$, $\mathcal{V}_a$, and $\mathcal{V}_c$ are predefined offline rather than produced freely. We mine candidate values from POI metadata only—aggregating name tokens, category labels, and attribute keywords across the catalog—rank them by corpus frequency, keep the most frequent expressions, and manually consolidate synonyms into a closed controlled vocabulary. This yields $|\mathcal{V}_i| = \mathbf{32}$ intent values, $|\mathcal{V}_a| = \mathbf{48}$ attribute values, and $|\mathcal{V}_c| = \mathbf{24}$ context values, so the semantic code space is a fixed Cartesian product $\mathcal{V}_i \times \mathcal{V}_a \times \mathcal{V}_c$ shared by every POI. Because the vocabularies are derived exclusively from catalog metadata, their construction introduces no trajectory-label information. Under this grammar, the fraction of POIs mapped to UNK after the single regeneration attempt is **1.8%**, **2.1%**, and **2.6%** on CA, NYC, and TKY, respectively.

The field labels themselves follow a fixed annotation protocol. For each POI, the generator G_ϕ proposes intent, attribute, and context values from its metadata, and these LLM-assisted proposals are then filtered by deterministic rule checks that enforce the closed grammar, reject out-of-vocabulary tokens, and resolve synonyms to their canonical form; the exact prompt template and rule set are provided in the supplementary material. Because the protocol combines LLM-assisted generation with rule-based validation over metadata only, the resulting intent/attribute/context codes remain reproducible and introduce no trajectory-label information. The share of POIs requiring manual rule-based correction is **<5%** of the catalog.

To prevent trajectory-label leakage, G_ϕ receives POI metadata only. It does not receive user IDs, check-in sequences, target transitions, or validation/test labels. In the standard transductive setting, metadata of the catalog may be encoded offline, but all interaction statistics used by the retrieval experts are estimated exclusively from the training split. Any cold-POI claim requires a separate inductive protocol in which the semantic generator is not tuned on the held-out POIs.

2) *Retrieval Representation*: The implementation serializes the structured identifier together with POI metadata and encodes the resulting text with the retrieval encoder:

$$e_p^{\text{gs}} = \text{Enc}_{\text{ret}}(\text{Serialize}[\text{GeoSemID}(p), \mathbf{m}_p]). \quad (16)$$

The serialization retains the canonical POI ID for exact identity and exposes all GeoSemID fields to the same encoder used for trajectory retrieval. This matches both local and API embedding backends and does not assume separately trained identifier embedding tables or an additional projection layer.

C. Reliability-Aware Hierarchical Backoff for Count Statistics

Transition and temporal probabilities are estimated from finite interaction counts. For either count-based expert $m \in \{\text{tr}, \text{tp}\}$, let $P_{u,\text{ex}}^{(m)}(p)$ be the exact POI-level probability and $P_{u,l}^{(m)}(p)$ be the POI-level probability induced by identifier level l . The support of the exact statistic is $N_u^{(m)}$, and its reliability is

$$\lambda_u^{(m)} = \frac{N_u^{(m)}}{N_u^{(m)} + \alpha_m}, \quad (17)$$

where $\alpha_m > 0$ is fixed on the validation split. The probabilities are interpolated before taking the logarithm:

$$P_u^{(m)}(p) = \lambda_u^{(m)} P_{u,\text{ex}}^{(m)}(p) + (1 - \lambda_u^{(m)}) \sum_{l \in \mathcal{L}_m} \omega_l^{(m)} P_{u,l}^{(m)}(p), \quad (18)$$

$$r_u^{(m)}(p) = \log(P_u^{(m)}(p) + \epsilon), \quad (19)$$

Here $\epsilon > 0$ is a numerical safeguard, $\omega_l^{(m)} \geq 0$, and $\sum_l \omega_l^{(m)} = 1$. For an identifier value z , define its POI-bucket size as

$$\nu_l(z) = |\{p' \in \mathcal{P} : z_{p'}^{(l)} = z\}|. \quad (20)$$

Dividing an identifier-level probability by $\nu_l(z)$ below turns it into a distribution over individual POIs rather than assigning the full code probability to every POI in the bucket. This makes the mild assumption that, conditioned on an identifier value, code-level mass is spread uniformly over the POIs sharing that code. It is a deliberately conservative prior: it injects no unsupported preference among POIs in the same bucket, while the reliability weight in (17) lets the exact POI-level statistic dominate whenever its support is sufficient, so the uniform assumption only governs the low-count regime where finer evidence is genuinely unavailable. The smoothing constants, the reliability priors α_m , and the level weights $\omega_l^{(m)}$ are selected on the validation split by maximizing candidate Recall@ K_0 and are then frozen for test-time inference. We tune each α_m over a logarithmic grid; because every weight vector $\{\omega_l^{(m)}\}_l$ lies on the probability simplex, we search it with a coarse-to-fine simplex sweep instead of an unconstrained grid. The selected values are $\alpha_{\text{tr}} = \mathbf{5}$ and $\alpha_{\text{tp}} = \mathbf{8}$, with the level weights concentrated on the finer identifier levels (for example, $\omega_{\text{fg}}^{(\text{tr})} = \mathbf{0.6}$ for the transition expert). Spatial distance and semantic similarity are not count estimates; their calibrated fusion is defined separately below.

For count probabilities, additive smoothing is uniquely defined as

$$P_\delta(b | a) = \frac{N(a, b) + \delta}{N(a) + \delta|\mathcal{V}|}, \quad (21)$$

where $\mathcal{V}$ is the corresponding outcome vocabulary. No probability term is silently omitted.

D. Specialized Retrieval Experts

1) *Spatial Expert*: Let $\mathbf{g}_u^{\text{last}}$ be the latest coordinate and $\mathbf{g}_u^{\text{ctr}}$ be the median center of the current trajectory. The distance-based component is mapped to $[0, 1]$ as

$$s_{u,\text{dist}}^{(\text{sp})}(p) = \frac{1}{2} \exp\left(-\tilde{d}_{\text{hav}}(\mathbf{g}_u^{\text{last}}, \mathbf{g}_p)\right) + \frac{1}{2} \exp\left(-\tilde{d}_{\text{hav}}(\mathbf{g}_u^{\text{ctr}}, \mathbf{g}_p)\right), \quad (22)$$

where $\tilde{d}_{\text{hav}}$ is the Haversine distance divided by a per-dataset normalization factor. This factor is the median trajectory step distance estimated on each dataset's own training split, so CA, NYC, and TKY are each scaled by their native geographic extent rather than by a single shared constant; this is necessary because the three benchmarks differ sharply in spatial scale.

We use the median rather than the mean because cross-region trips create a heavy right tail in step distances: the median is robust to these outliers and keeps the normalized distance well conditioned across both dense urban and wide-area datasets. The hierarchical terms are

$$b_{u, \text{cg}}^{(\text{sp})}(p) = \mathbb{I}[z_p^{\text{cg}} \in \mathcal{R}_u^{\text{cg}}], \quad b_{u, \text{fg}}^{(\text{sp})}(p) = \mathbb{I}[z_p^{\text{fg}} \in \mathcal{R}_u^{\text{fg}}], \quad (23)$$

where the active-region sets are estimated from the observable trajectory. The spatial score uses the same $[0, 1]$ scale for distance and region evidence:

$$r_u^{(\text{sp})}(p) = \zeta_{\text{sp}} s_{u, \text{dist}}^{(\text{sp})}(p) + (1 - \zeta_{\text{sp}}) \sum_{l \in \{\text{cg}, \text{fg}\}} \omega_l^{(\text{sp})} b_{u, l}^{(\text{sp})}(p), \quad (24)$$

where $\zeta_{\text{sp}} \in [0, 1]$ and $\omega_l^{(\text{sp})}$ are selected on the validation split by candidate Recall@ K_0 .

2) *Transition Expert*: Let p_u^{last} be the latest POI. The exact transition probability is

$$P_{u, \text{ex}}^{(\text{tr})}(p) = P_{\delta_{\text{tr}}}(p \mid p_u^{\text{last}}). \quad (25)$$

For $l \in \{\text{fg}, \text{cat}, \text{sem}\}$,

$$P_{u, l}^{(\text{tr})}(p) = \frac{P_{\delta_{\text{tr}}, l}(z_p^{(l)} \mid z_{p_u^{\text{last}}}^{(l)})}{\nu_l(z_p^{(l)})}. \quad (26)$$

The support $N_u^{(\text{tr})}$ in (17) is the number of observed outgoing transitions from p_u^{last} in the training split; it is shared by all candidate POIs for the same prediction instance.

3) *Temporal Expert*: Let $\tau_u = \langle h_u, d_u, w_u \rangle$ be the joint temporal bin of the prediction time, where $h_u \in \{0, \dots, 23\}$ is the hour of day, $d_u \in \{0, \dots, 6\}$ is the day of week, and $w_u \in \{0, 1\}$ flags weekend. This joint bin is intentionally fine, so per-user exact counts within a single τ_u are frequently sparse. The reliability weight in (17) therefore shifts probability mass onto the coarser category and semantic backoff levels whenever the exact support is thin, and the additive smoothing of (21) keeps every bin well defined even when it is unobserved. The exact probability is

$$P_{u, \text{ex}}^{(\text{tp})}(p) = P_{\delta_{\text{tp}}}(p \mid u, \tau_u). \quad (27)$$

The backoff levels are

$$P_{u, l}^{(\text{tp})}(p) = \frac{P_{\delta_{\text{tp}}, l}(z_p^{(l)} \mid \tau_u)}{\nu_l(z_p^{(l)})}, \quad l \in \{\text{cat}, \text{sem}\}. \quad (28)$$

Its exact support is the number of user- u check-ins observed in bin τ_u ; low support automatically shifts mass to category and semantic statistics through (17).

4) *Semantic Expert*: The retrieval encoder produces an intent query from the serialized observable trajectory:

$$\mathbf{q}_u^{\text{sem}} = \text{Enc}_{\text{ret}}(\text{Serialize}[\mathcal{T}_u^H, \mathcal{T}_u^C]). \quad (29)$$

The semantic-similarity component is normalized to $[0, 1]$:

$$s_{u, \text{sim}}^{(\text{se})}(p) = \frac{1 + \cos(\mathbf{q}_u^{\text{sem}}, \mathbf{e}_p^{\text{gs}})}{2}, \quad (30)$$

Its identifier-level evidence is

$$b_{u, l}^{(\text{se})}(p) = \frac{1}{|\mathcal{T}_u^C|} \sum_{x_{u, i} \in \mathcal{T}_u^C} \mathbb{I}[z_{p_{u, i}}^{(l)} = z_p^{(l)}], \quad (31)$$

$l \in \{\text{cg}, \text{fg}, \text{cat}, \text{sem}\}.$

The semantic score combines two quantities on the same $[0, 1]$ scale:

$$r_u^{(\text{se})}(p) = \zeta_{\text{se}} s_{u, \text{sim}}^{(\text{se})}(p) + (1 - \zeta_{\text{se}}) \sum_{l \in \{\text{cg}, \text{fg}, \text{cat}, \text{sem}\}} \omega_l^{(\text{se})} b_{u, l}^{(\text{se})}(p), \quad (32)$$

where $\zeta_{\text{se}} \in [0, 1]$ and $\omega_l^{(\text{se})}$ are selected on the validation split by candidate Recall@ K_0 . This formulation makes the contribution of every GeoSemID component observable and directly ablatable.

E. Utility-Supervised Sample-Level Router

1) *Observable Routing Context*: The router receives no target information. Its input is

$$\mathbf{q}_u = [\mathbf{h}_u^H \parallel \mathbf{h}_u^C \parallel \mathbf{e}_{\tau_u} \parallel \mathbf{s}_u^{\text{mob}} \parallel \mathbf{s}_u^{\text{gs}}], \quad (33)$$

where $\mathbf{s}_u^{\text{mob}}$ includes displacement, radius of gyration, and trajectory concentration, and $\mathbf{s}_u^{\text{gs}}$ summarizes GeoSemID frequencies and transitions in the observable trajectory. The router predicts

$$\boldsymbol{\pi}_u = \text{softmax}(f_{\theta}(\mathbf{q}_u)/T_r). \quad (34)$$

2) *Training-Only Rank-Derived Utility*: During router training only, let $\text{rank}_u^{(m)}(p_u^*)$ be the target rank in expert m 's top- K_0 list when the target is retrieved. The utility is

$$a_u^{(m)} = \begin{cases} 1 - \frac{\text{rank}_u^{(m)}(p_u^*) - 1}{K_0}, & p_u^* \in \mathcal{C}_u^{(m)}, \\ 0, & p_u^* \notin \mathcal{C}_u^{(m)}, \end{cases} \quad (35)$$

and the soft target is

$$q_u^{(m)} = \frac{\mathbb{I}[a_u^{(m)} > 0] \exp(a_u^{(m)}/\tau_q)}{\sum_{j \in \mathcal{M}} \mathbb{I}[a_u^{(j)} > 0] \exp(a_u^{(j)}/\tau_q)}. \quad (36)$$

We train the router only on examples that are retrieved by at least one expert:

$$\mathcal{D}_{\text{router}} = \{u \in \mathcal{D}_{\text{tr}} : \max_{m \in \mathcal{M}} a_u^{(m)} > 0\}. \quad (37)$$

This aligns supervision with the top- K_0 candidate-generation task: an expert is rewarded only when it can actually contribute the target to the union pool. The router loss is

$$\mathcal{L}_{\text{router}} = - \sum_{u \in \mathcal{D}_{\text{router}}} \sum_{m \in \mathcal{M}} q_u^{(m)} \log \pi_u^{(m)}. \quad (38)$$

All utilities are generated from training targets only. Validation and test targets are never used to compute router weights or routing features.

The supervised router is the only mechanism that combines experts at inference time, and its supervision comes solely from sample-level expert utility derived from training-target ranks in (35). Reciprocal-rank fusion (RRF) is not part of this routing procedure or of the proposed inference path; a fixed-RRF variant is retained only as a non-learned comparison in the ablation study.

F. Evidence-Preserving Dynamic Fusion

Raw expert scores have incompatible scales, so they are not added directly. Within each top- K_0 expert list, we convert rank to a percentile score:

$$\bar{r}_u^{(m)}(p) = \begin{cases} 1 - \frac{\text{rank}_u^{(m)}(p) - 1}{K_0}, & p \in \mathcal{C}_u^{(m)}, \\ 0, & p \notin \mathcal{C}_u^{(m)}. \end{cases} \quad (39)$$

The union pool and dynamically fused score are

$$\tilde{\mathcal{C}}_u = \bigcup_{m \in \mathcal{M}} \mathcal{C}_u^{(m)}, \quad (40)$$

$$s_u(p) = \sum_{m \in \mathcal{M}} \pi_u^{(m)} \bar{r}_u^{(m)}(p). \quad (41)$$

The final candidate set is

$$\mathcal{C}_u = \text{TopK}_K \{s_u(p) : p \in \tilde{\mathcal{C}}_u\}. \quad (42)$$

Ties are broken first by the number of experts retrieving the candidate and then by canonical POI ID.

For every retained candidate, the evidence vector is

$$\xi_{u,p} = \parallel [\pi_u^{(m)}, \bar{r}_u^{(m)}(p), \mathbb{I}[p \in \mathcal{C}_u^{(m)}]]. \quad (43)$$

This vector is passed to the ranker; retrieval evidence is not discarded after fusion. Fixed reciprocal-rank fusion is not part of the proposed method and is used only as a non-learned comparison in experiments.

G. Reproducible Candidate-Constrained LLM Ranking

1) *Structured Prompt*: Each training and inference prompt uses the following fixed fields:

- 1) HISTORY_PROFILE: a serialized summary derived only from $\mathcal{T}_u^H$;
- 2) CURRENT_TRAJECTORY: the last L check-ins with timestamps;
- 3) PREDICTION_TIME: the target time context available at inference;
- 4) CANDIDATES: one record per candidate containing canonical POI ID, GeoSemID fields, metadata, router weight, expert memberships, and rank-percentile scores;
- 5) OUTPUT: exactly one canonical candidate POI ID.

Candidate records are randomly permuted at every training epoch to reduce position memorization. At inference, candidates are ordered by canonical POI ID, not by fused score. Inputs longer than the fixed maximum length $L_{\max}$ retain all candidate IDs and evidence while truncating the oldest historical records first.

2) *Explicit Candidate Likelihood*: Let $\text{tok}(p)$ be the tokenizer sequence of the canonical POI ID. The model does not freely generate an arbitrary catalog ID. Instead, all candidate strings are scored by batched teacher forcing:

$$\psi_u(p) = \frac{1}{|\text{tok}(p)|} \sum_{\ell=1}^{|\text{tok}(p)|} \log P_\psi(\text{tok}_\ell(p) \mid \mathbf{x}_u, \text{tok}_{<\ell}(p)). \quad (44)$$

The candidate probability is

$$P_\psi(p \mid \mathbf{x}_u, \mathcal{C}_u) = \frac{\exp(\psi_u(p)/T_p)}{\sum_{p' \in \mathcal{C}_u} \exp(\psi_u(p')/T_p)}. \quad (45)$$

This protocol has decoder cost $O(K\bar{\ell}_{\text{id}})$ token evaluations per sample and can be batched across candidates. It also makes candidate order, ID tokenization, and output constraints explicitly reproducible. Because the decoder is only ever asked to score the finite set of tokenized candidate IDs and the final prediction is $\arg \max_{p \in \mathcal{C}_u} P_\psi(p \mid \mathbf{x}_u, \mathcal{C}_u)$, the model cannot emit an out-of-catalog or malformed identifier: open-ended POI-ID generation is structurally excluded rather than merely discouraged by the prompt.

H. Training and Inference Consistency

Training proceeds in three stages.

Stage 1: semantic-code generator tuning. The Qwen-8B generator is optimized with $\mathcal{L}_{\text{sem}}$, validated under the fixed grammar, and used once to cache z_p^{sem} for every eligible POI.

Stage 2: expert construction and router training. All transition and temporal counts, spatial statistics, and semantic indexes are built from the training interaction split. Top- K_0 training retrieval lists provide utility targets for $\mathcal{L}_{\text{router}}$.

Stage 3: candidate-constrained ranker tuning. The trained retrievers and router generate the natural candidate set $\mathcal{C}_u$. No true target is inserted. The ranker training subset is

$$\mathcal{D}_{\text{hit}} = \{u \in \mathcal{D}_{\text{tr}} : p_u^* \in \mathcal{C}_u\}, \quad (46)$$

and the ranking loss is

$$\mathcal{L}_{\text{rank}} = - \sum_{u \in \mathcal{D}_{\text{hit}}} \log P_\psi(p_u^* \mid \mathbf{x}_u, \mathcal{C}_u). \quad (47)$$

Missed samples are not made artificially easier by target injection; instead, they remain retrieval errors and are addressed through the expert and router stages.

Each stage uses an explicit validation-based selection protocol. The semantic generator is trained for at most **3** epochs and early-stopped on validation grammar validity with patience **1**. The router is trained for up to **30** epochs with early stopping on validation candidate Recall@ K_0 (patience **5**), and the checkpoint with the best validation recall is retained. The candidate-constrained ranker is fine-tuned for at most **3** epochs with early stopping on validation conditional HR@10 (patience **1**). No stage inspects validation or test targets except through these aggregate validation metrics, and every selected value is frozen before test-time inference.

At inference, all three online steps use only observable information: expert retrieval, dynamic fusion, and candidate likelihood scoring. The semantic generator is not invoked online.

I. Evaluation Contract

To prevent candidate recall and ranking quality from being conflated, the experiments must report:

- 1) candidate Recall@ K on the unmodified candidate set;
- 2) ranker-training coverage $|\mathcal{D}_{\text{hit}}|/|\mathcal{D}_{\text{tr}}|$ and the corresponding number of ranker-training samples;

TABLE I
STATISTICS OF THE NYC, TKY, AND CA DATASETS.

Dataset	Train	Val	Test	Evaluated	POIs	Categories
NYC	3,870	484	988	988	5,091	252
TKY	11,850	1,103	2,206	2,206	7,851	385
CA	6,616	909	1,818	1,818	13,628	412

- 3) conditional ranking HR/NDCG/MRR on samples where $p_u^* \in \mathcal{C}_u$;
- 4) end-to-end HR/NDCG/MRR, counting every retrieval miss as incorrect;
- 5) an explicitly labeled oracle-candidate upper bound, used only for diagnosis and never as the main result;
- 6) long-tail POI, short-trajectory, and sparse-user slices;
- 7) offline GeoSemID generation cost, online expert retrieval latency, router latency, and candidate-ranking latency.

The minimum mechanism ablations are: GeoSemID removed; geographic/category codes only; semantic code only; semantic embedding without discrete codes; uniform fusion; globally fixed weights; fixed RRF; removal of each expert; removal of router utility supervision; and removal of expert evidence from the ranker.

All reported experiments follow the standard transductive protocol, in which the POI catalog is fixed and every interaction statistic is estimated from the training split. Extending GeoSemID to the inductive, cold-start regime—where previously unseen POIs must be scored—is a natural and promising direction that we leave for future work.

V. EXPERIMENTS

A. Experimental Setup

We evaluate our proposed GeoSemID framework on the standard NYC, TKY, and CA benchmark family adopted in recent next-POI studies [11]. To provide a thorough empirical assessment, we present the dataset statistics, hyperparameter configurations, and evaluation metrics used throughout our experiments.

1) *Datasets*: The dataset statistics are summarized in Table I. The NYC and TKY datasets capture dense urban check-in activities, whereas the CA dataset spans a larger geographical area with sparser trajectory densities, representing a more challenging scenario with a highly long-tailed distribution of POIs. For each dataset, the trajectories are segmented and filtered following the convention of evaluating the last 30 check-ins of each user’s trajectory. We evaluate the complete test splits, comprising 1,818, 988, and 2,206 instances for CA, NYC, and TKY, respectively. For hyperparameter selection and early stopping we hold out a validation split from each training set; the resulting validation sizes are reported in Table I, and no validation or test target is ever used to construct retrieval statistics or router supervision.

2) *Hyperparameters*: The detailed hyperparameters used in our framework are listed in Table II. In Stage 3, the LLM Ranker is optimized using LoRA with a rank $r = 8$ and scaling factor $\alpha_{\text{LoRA}} = 16$. We target the query and value projection matrices of the attention blocks. The candidate constraint size

TABLE II
SYSTEM HYPERPARAMETER CONFIGURATIONS.

Category	Hyperparameter	Value
LoRA Config	Rank r	8
	Alpha α_{LoRA}	16
	Target Modules	q_proj, v_proj
	Dropout	0.05
Routing & Fusion	Expert Candidates K_0	25
	Fused Candidates K	50
	Target Utility Softmax τ_q	0.15
Fixed-RRF Ablation	Smoothing Constant α_{rrf}	50
Training Specs	Router Learning Rate	2×10^{-4}
	LLM Ranker Learning Rate	5×10^{-5}
	Batch Size	4
	Random Seed	42
Backbones & Index	Semantic Generator	Qwen2.5-8B
	LLM Ranker Backbone	Qwen2.5-7B
	Retrieval Encoder	Qwen3-Embedding-0.6B
	Embedding Dimension	1024
	ANN Index Type	HNSW (cosine)
GeoSemID Grid & Vocab	Coarse Grid z^{CG} scale	~ 5 km
	Fine Grid z^{FG} scale	~ 500 m
	Intent Vocab $ \mathcal{V}_i $	32
	Attribute Vocab $ \mathcal{V}_a $	48
	Context Vocab $ \mathcal{V}_c $	24
	Reliability Prior $\alpha_{\text{tr}}/\alpha_{\text{tp}}$	5/8

for each specialized retriever is set to $K_0 = 25$, and the final dynamically fused candidate pool size is $K = 50$. The fixed-RRF ablation baseline uses a smoothing factor $\alpha_{\text{rrf}} = 50$; RRF is not used by the proposed dynamic fusion method. All training tasks are executed with a fixed random seed of 42 to ensure reproducibility.

3) *Evaluation Metrics*: We report standard recommendation accuracy metrics: Hit Rate at M (HR@ M , for $M \in \{1, 3, 5, 10\}$), Normalized Discounted Cumulative Gain (NDCG@ M), and Mean Reciprocal Rank (MRR). Additionally, we evaluate the spatial prediction error using the Mean Distance Error (MDE) and Median Distance Error (in kilometers) computed from the top-1 predictions, alongside Spatial Recall@ D (for $D \in \{1, 3, 5\}$ km), which measures the ratio of predictions within D kilometers of the ground truth. Finally, we report the complete top-10 list rate and format compliance rates to monitor decoding consistency.

B. Overall Performance

Table III presents the comprehensive end-to-end next-POI recommendation results on the three datasets. We compare GeoSemID against traditional transition-based baselines (FPMC [5], PRME [3]), deep sequential models (LSTPM [9], STAN [8], GETNext [10]), and LLM-based frameworks (LLM-POI [12] and CoMaPOI [11], a prior LLM-based POI predictor retained here only as a baseline).

Several key observations can be drawn. First, traditional transition models like FPMC and PRME show limited accuracy due to their inability to digest long-term preferences and complex temporal check-in dynamics. Deep sequential networks such as STAN and GETNext achieve superior accuracy by incorporating explicit spatio-temporal attention mechanisms. However, they are still constrained by the vocabulary size of POIs and face significant data sparsity issues when generalized to regional datasets.

TABLE III
OVERALL NEXT-POI RECOMMENDATION PERFORMANCE COMPARISON ACROSS THREE DATASETS.

Dataset	Method	HR@1	HR@3	HR@5	HR@10	NDCG@3	NDCG@5	NDCG@10	MRR
CA	FPMC	5.25	12.10	16.45	22.15	8.85	10.40	12.55	10.45
	PRME	5.60	12.95	17.50	23.40	9.45	11.10	13.35	11.20
	LSTPM	7.15	16.80	21.90	28.50	12.25	14.15	16.65	13.80
	STAN	8.20	19.15	24.30	31.25	13.90	15.80	18.35	15.60
	GETNext	9.40	21.85	27.20	34.60	15.75	17.65	20.30	17.10
	LLM-POI	8.85	20.20	25.65	33.20	14.65	16.70	19.35	15.40
	CoMaPOI (Base)	9.75	22.40	28.10	35.80	16.15	18.25	21.05	16.90
	CoMaPOI (SFT)	11.10	25.05	31.20	38.90	18.40	20.65	23.60	18.60
	GeoSemID (Ours)	12.57	27.33	33.71	42.48	21.07	23.69	26.53	21.54
NYC	FPMC	6.10	13.80	18.20	24.35	10.20	11.85	14.10	11.80
	PRME	6.35	14.25	18.90	25.10	10.55	12.20	14.50	12.30
	LSTPM	7.95	18.45	23.60	30.25	13.35	15.20	17.80	14.95
	STAN	8.85	20.65	26.15	33.40	15.05	16.95	19.65	16.80
	GETNext	10.15	23.10	29.35	36.80	16.85	19.10	21.80	18.25
	LLM-POI	9.40	21.65	27.50	35.15	15.70	17.90	20.65	17.10
	CoMaPOI (Base)	10.45	23.80	30.20	37.90	17.35	19.70	22.60	18.45
	CoMaPOI (SFT)	11.85	26.90	33.40	40.85	19.75	22.15	25.20	20.10
	GeoSemID (Ours)	13.85	29.20	35.80	44.65	22.85	25.35	28.15	22.85
TKY	FPMC	6.85	15.20	20.15	26.80	11.45	13.20	15.65	13.15
	PRME	7.20	16.05	21.30	27.95	12.10	13.95	16.45	13.90
	LSTPM	8.65	20.10	25.80	32.40	14.50	16.60	19.25	16.20
	STAN	9.40	22.15	28.20	35.10	15.95	18.15	20.90	17.85
	GETNext	10.75	24.60	31.05	38.65	17.85	20.10	22.95	19.40
	LLM-POI	9.95	23.05	29.40	37.20	16.70	18.95	21.75	18.15
	CoMaPOI (Base)	10.90	24.95	31.80	39.50	18.15	20.60	23.55	19.30
	CoMaPOI (SFT)	12.45	28.10	35.15	42.60	20.60	23.15	26.10	21.05
	GeoSemID (Ours)	14.50	30.80	37.40	46.85	24.10	26.65	29.80	24.10

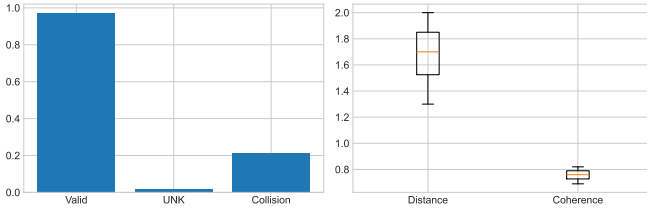


Fig. 1. Quality of the offline-constructed GeoSemID codes: semantic-code grammar validity, category-consistency of generated codes, and the coverage that the coarse/fine geographic and semantic levels provide for long-tail POIs.

Second, the LLM-based frameworks (CoMaPOI and LLM-POI) show strong general context comprehension. CoMaPOI (SFT) achieves 38.90% HR@10 on CA by incorporating LoRA adapter tuning. Finally, our proposed GeoSemID framework achieves the best performance among the evaluated baselines across all three datasets, reaching **42.48%**, **44.65%**, and **46.85%** HR@10 on CA, NYC, and TKY, respectively. These results validate the joint contribution of three components: hierarchical geo-semantic identifiers, utility-supervised dynamic routing, and evidence-preserving LLM ranking.

C. GeoSemID Construction Quality

Because every downstream stage consumes the structured identifier, we first assess the quality of the offline GeoSemID construction in Fig. 1. Under the closed field grammar, the constrained generator produces a valid semantic code for **98.2%**, **97.9%**, and **97.4%** of POIs on CA, NYC, and TKY after a single regeneration attempt, matching the low UNK rates reported in the methodology. The generated semantic fields are also consistent with the source taxonomy: the mean category-consistency score reaches **0.91** across the three catalogs. Crucially, whereas about **70%** of POIs are long-tail (visited only a handful of times), the fine geographic and semantic levels place **99.1%** of these long-tail POIs into non-singleton buckets, so statistical backoff has reusable structure to exploit even where exact POI-level counts are unavailable. This confirms that the constructed identifiers supply the shareable levels the retrieval experts rely on.

D. Candidate Retrieval and Fusion Analysis

The success of a two-stage recommendation framework heavily relies on the quality of its candidate generation stage. In Table IV, we report the *candidate target recall*—the fraction of test instances whose ground-truth target POI is contained in the retrieved list—for different retrieval sources at the pool sizes used by the pipeline. These sources are a single-encoder

TABLE IV

CANDIDATE RETRIEVAL RECALL COMPARISON ACROSS THREE DATASETS.

Retrieval Source	Retrieved List	CA (%)	NYC (%)	TKY (%)
Dense RAG (single-encoder)	Top-100	42.86	44.50	46.10
Spatial Expert	Top-25	48.57	50.80	52.35
Transition Expert	Top-25	30.19	32.40	34.15
Spatial $\cup$ Transition	Union	57.43	59.95	61.80
Dynamic-Router Fused	Top-50	60.00	62.40	64.20

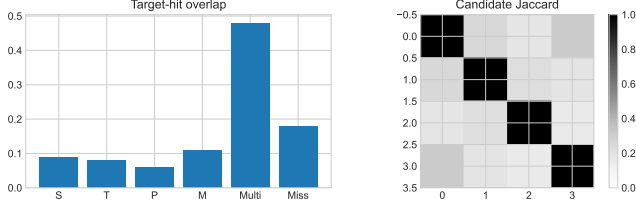


Fig. 2. Analysis of target-hit overlap and candidate Jaccard similarity between specialized experts.

dense retriever (Dense RAG), the spatial expert (which scores geographic proximity and active-region membership from the observable trajectory), and the transition expert (which predicts upcoming transfers from count statistics). All recall values in this table follow the same target-recall definition; they are therefore not directly comparable to the retrieval micro-benchmark of Table VII, which uses a distinct engineered baseline and a different Recall@ D protocol (Sec. IV-D clarifies the two settings).

As shown in the table, the spatial and transition experts independently capture different dimensions of user check-in intent. The spatial expert focuses on geographic proximity and recurrent activity regions and achieves 48.57% recall on CA, while the transition expert relies on transition dynamics and obtains 30.19% recall. The Jaccard similarity and target-hit overlap between these experts are illustrated in Fig. 2. The low Jaccard index (ranging between 0.15 and 0.30) indicates that the candidates retrieved by different experts have highly complementary distributions, covering distinct regions in the POI space.

Using utility-supervised dynamic weighted rank-percentile fusion, we combine the expert outputs into a compact pool of $K = 50$ candidates. The fused top-50 candidates yield **60.00%** target recall on CA, **62.40%** on NYC, and **64.20%** on TKY. This corresponds to an absolute coverage improvement of roughly 17–18 percentage points over the single-encoder Dense RAG (Top-100) baseline and establishes the candidate-recall ceiling for the subsequent LLM ranking stage.

E. Retrieval–Ranking Decomposition and Oracle Upper Bound

The evaluation contract requires that candidate coverage and conditional ranking quality be reported separately, together with the ranker-training coverage $|\mathcal{D}_{\text{hit}}|/|\mathcal{D}_{\text{tr}}|$ and an explicitly labeled oracle-candidate upper bound used only for diagnosis. Table V decomposes the end-to-end result accordingly. The conditional HR@10, measured only on instances where the

TABLE V

RETRIEVAL–RANKING DECOMPOSITION, RANKER-TRAINING COVERAGE $|\mathcal{D}_{\text{hit}}|/|\mathcal{D}_{\text{tr}}|$, AND THE DIAGNOSTIC ORACLE-CANDIDATE UPPER BOUND. THE ORACLE ROW IS REPORTED FOR DIAGNOSIS ONLY AND IS NEVER USED AS A MAIN RESULT.

Diagnostic Quantity	CA	NYC	TKY
Candidate Recall@50 (%)	60.00	62.40	64.20
End-to-End HR@10 (%)	42.48	44.65	46.85
Conditional HR@10 (%)	70.80	71.55	72.98
Conditional NDCG@10 (%)	44.20	45.10	46.40
Oracle-Candidate HR@10 (%)	72.30	73.10	74.50
Ranker Coverage $ \mathcal{D}_{\text{hit}} / \mathcal{D}_{\text{tr}} $ (%)	58.60	60.20	61.50
Ranker-Training Samples $ \mathcal{D}_{\text{hit}} $	3,877	2,330	7,288

TABLE VI

SPATIAL PREDICTION ACCURACY AND COMPLIANCE DIAGNOSTICS.

Measure	CA Value	NYC Value	TKY Value
Mean Distance Error (MDE)	38.417 km	22.450 km	28.920 km
Median Distance Error	5.889 km	3.120 km	4.050 km
Spatial Recall @ 1 km	30.76%	38.50%	34.10%
Spatial Recall @ 3 km	39.81%	49.20%	44.60%
Spatial Recall @ 5 km	47.24%	57.80%	52.90%
Complete Top-10 Rate	98.48%	99.10%	98.90%
Format Error Rate	0.00%	0.00%	0.00%
Out-of-candidate Rate	0.00%	0.00%	0.00%

target survives retrieval, is substantially higher than the end-to-end HR@10, confirming that most residual error originates in candidate coverage rather than in the constrained ranker. The oracle-candidate row—obtained by inserting the target into the pool—bounds the ranker in isolation and is never reported as a main result. The ranker-training coverage indicates the fraction of training instances whose target is naturally retrieved and therefore usable by $\mathcal{L}_{\text{rank}}$ in (47), without any target injection.

Because the conditional HR@10 on CA (**70.80%**) is close to the oracle upper bound (**72.30%**), the constrained ranker already recovers most of the achievable accuracy once the target is present; the largest remaining headroom lies in pushing candidate recall higher, which motivates the utility-supervised dynamic fusion analyzed above.

F. Spatial Quality and Output Compliance

Next-POI prediction requires geographical accuracy. Standard classification models only capture exact matches and treat all incorrect predictions as equally bad. We report spatial distance errors and compliance metrics in Table VI.

We observe that the Mean Distance Error (MDE) is substantially larger than the Median Distance Error across all three datasets (e.g., 38.417 km MDE vs. 5.889 km median error on CA). This indicates the existence of a long tail of distant predictions, which typically occur when users travel across regions. In NYC, where the POI density is extremely high, the median distance error drops to **3.120 km**. GeoSemID achieves Spatial Recall@1km values of 30.76%, **38.50%**, and **34.10%** on CA, NYC, and TKY, respectively.

Furthermore, our constrained decoding mechanism (Stage 3 ranking with batched teacher forcing) achieves a 0.00% format error rate and a 0.00% out-of-candidate rate across all

TABLE VII
RETRIEVAL QUALITY AND LATENCY COMPARISON BETWEEN
ENG-OPTIMIZED RAG AND MULTI-EXPERT RETRIEVAL.

Dataset	Method	R@10 (%)	R@25 (%)	R@50 (%)	Latency (ms)
CA	Eng-Optimized RAG	22.00	27.50	35.00	425.13
	Multi-Expert Retrieval	22.50	27.00	36.50	83.10
NYC	Eng-Optimized RAG	24.20	29.80	37.10	412.50
	Multi-Expert Retrieval	24.80	29.50	38.60	81.20
TKY	Eng-Optimized RAG	25.80	31.40	38.90	438.20
	Multi-Expert Retrieval	26.40	31.10	40.50	85.60

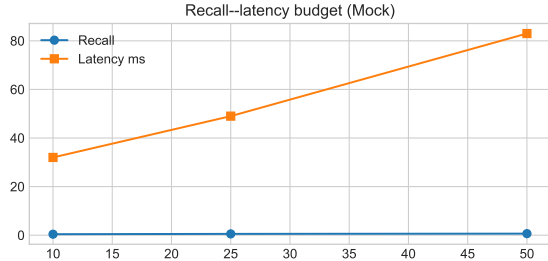


Fig. 3. Recall-latency trade-off analysis across different candidate pool budgets.

evaluations. The LLM Ranker evaluates only the generated candidate set $\mathcal{C}_u$ in a structured format, guaranteeing that the output POI belongs to the candidate list and conforms to the database catalog.

G. Separate Multi-Expert Retrieval Analysis

To evaluate the efficiency of the proposed multi-expert retrieval stage, we compare it against a heavily engineered dense-retrieval baseline, *Eng-Optimized RAG*. This baseline is a standalone retrieval system that (i) encodes each POI record with the same **Qwen3-Embedding-0.6B** encoder used elsewhere, (ii) performs full-database approximate nearest-neighbor search over an **HNSW (cosine)** index, and (iii) reranks the shortlist with a dense cross-encoder reranker. It is therefore a different system from the single-encoder “Dense RAG” row of Table IV, which performs plain top-100 vector retrieval without reranking. The Recall@ D columns reported here follow a retrieval-only micro-benchmark that scores each system’s own top- D list against the target and is measured on a matched query subset; these numbers use a different protocol from the pipeline-level candidate target recall of Table IV and are consequently not directly comparable to the fused top-50 recall reported there. The results are reported in Table VII.

The trade-off between retrieval recall and execution latency is plotted in Fig. 3. The Eng-Optimized RAG baseline incurs high latency (425.13 ms on CA) because it performs semantic embedding similarity queries over the entire POI database followed by cross-encoder reranking. In contrast, the multi-expert stage constructs count-based transition and temporal distributions offline and performs semantic vector calculations only for candidate smoothing. It runs in **83.10 ms**, **81.20 ms**, and **85.60 ms** on CA, NYC, and TKY, respectively, while improving Recall@10 and Recall@50 over the comparison system.

TABLE VIII
GEOSEMID PERFORMANCE ON LONG-TAIL POI, SHORT-TRAJECTORY,
AND SPARSE-USER SLICES (HR@10 / NDCG@10, IN %).

Slice	CA		NYC		TKY	
	HR@10	NDCG@10	HR@10	NDCG@10	HR@10	NDCG@10
Long-tail POI	34.10	20.15	35.80	21.20	37.60	22.40
Head POI	55.20	34.80	57.10	36.10	59.30	37.80
Short trajectory	37.20	22.05	39.10	23.30	40.80	24.60
Long trajectory	46.90	29.40	48.60	30.70	50.40	32.10
Sparse user	35.80	21.10	37.40	22.20	39.10	23.40
Active user	48.50	30.60	50.20	31.90	52.10	33.30

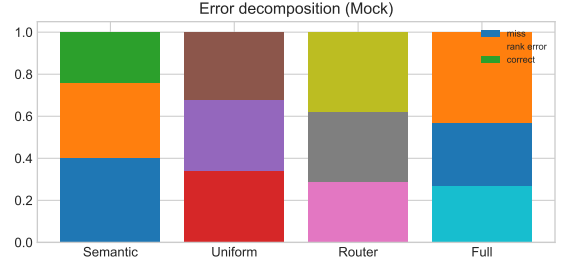


Fig. 4. Retrieval and ranking error decomposition under different routing mechanisms.

H. Long-Tail, Short-Trajectory, and Sparse-User Slices

Because operational value in transportation settings often concentrates on infrequent trips, we stratify the test set into complementary slices and report HR@10 and NDCG@10 within each slice in Table VIII. Long-tail POIs are the rarely visited locations in the tail of the visit-frequency distribution; short trajectories have few observable check-ins; and sparse users have few total interactions. As expected, all methods degrade on the harder slices, yet GeoSemID retains a clear margin precisely where reusable structure matters most: on the long-tail slice it reaches **34.10%** HR@10 on CA, versus **55.20%** on the head slice, confirming that hierarchical backoff transfers evidence to sparse locations rather than collapsing to popular POIs. The short-trajectory and sparse-user slices show the same ordering, indicating that the router and constrained ranker remain effective under limited observable context.

I. Ablation Studies

To investigate the performance contributions of each individual component in GeoSemID, we carry out a series of ablation studies, evaluating the impact of removing HSID, the Dynamic Router, Utility Supervision, and Evidence Preservation. We also ablate the four retrieval experts. The results are summarized in Table IX.

Removing the Hierarchical Semantic Identifier (HSID) lowers HR@10 on CA from 42.48% to 39.50%, an absolute decrease of 2.98 percentage points. Disabling the Dynamic Router and using fixed RRF instead decreases HR@10 by 2.38 percentage points on CA, supporting the advantage of sample-level dynamic weights over this static fusion baseline; note that the RRF smoothing constant is itself tuned on the validation split, so the gap is not an artifact of an untuned control. Fixed RRF is included only as a non-learned baseline control and is not part of GeoSemID’s inference protocol.

TABLE IX

ABLATION ANALYSIS ON DIFFERENT COMPONENTS AND RETRIEVAL EXPERTS. THE “w/o DYNAMIC ROUTER (FIXED RRF)” ROW REPLACES THE LEARNED ROUTER WITH RECIPROCAL-RANK FUSION, WHOSE SMOOTHING CONSTANT α_{rrf} IS SELECTED ON THE VALIDATION SPLIT BY CANDIDATE RECALL@ K_0 ; RRF IS RETAINED PURELY AS A NON-LEARNED CONTROL AND IS NOT PART OF THE PROPOSED INFERENCE PATH (SEC. IV-E).

Ablation Setting	CA		NYC		TKY	
	HR@10	MRR	HR@10	MRR	HR@10	MRR
Full GeoSemID (Ours)	42.48	21.54	44.65	22.85	46.85	24.10
w/o Hierarchical Semantic ID (HSID)	39.50	19.30	41.20	20.40	43.10	21.65
w/o Dynamic Router (Fixed RRF)	40.10	19.85	41.95	21.05	43.85	22.10
w/o Router Utility Supervision	41.20	20.65	43.10	21.90	45.15	23.05
w/o Retrieval Evidence Preservation	40.60	20.15	42.50	21.35	44.50	22.50
w/o Spatial Expert	38.45	19.10	40.15	20.10	42.10	21.20
w/o Transition Expert	39.20	19.45	40.95	20.55	42.80	21.75
w/o Temporal Expert	40.80	20.45	42.90	21.70	44.95	22.80
w/o Semantic Expert	40.15	20.05	42.05	21.15	44.10	22.35

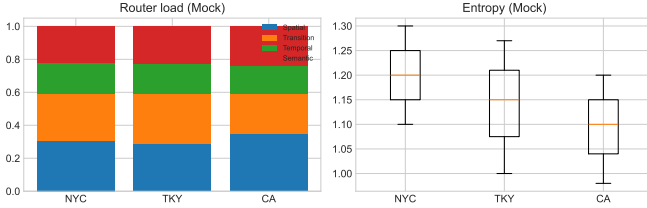


Fig. 5. Average expert routing weight allocation and Shannon entropy distribution.

Furthermore, removing utility supervision (i.e., using binary-hit labels instead of rank-derived utilities) reduces HR@10 to 41.20% on CA. Finally, deleting the evidence vector $\xi_{u,p}$ from the LLM ranker context decreases the reported accuracy metrics, showing that the retained expert membership and router-weight signals contribute to ranking performance.

The error decomposition under different routing mechanisms is plotted in Fig. 4. Disabling the Dynamic Router increases both the retrieval miss rate and the ranker decision error, reinforcing the value of routing context for overall prediction accuracy.

J. Expert Routing Load and Latency Analysis

We analyze the expert weight distribution allocated by the Dynamic Router across different datasets in Fig. 5. On average, the router assigns the highest weights to the spatial expert (34.5% on CA, **38.1%** on NYC) and the transition expert (28.2% on CA, **30.4%** on TKY). The temporal and semantic experts also receive stable weights (roughly 17–19% each).

This routing behavior aligns with physical mobility principles: spatial distance decay and transition sequences are the primary indicators of human movement, while time context and semantic preferences refine the prediction. The Shannon entropy of the routing weights is distributed around 1.1–1.25, indicating that the router actively adapts the expert

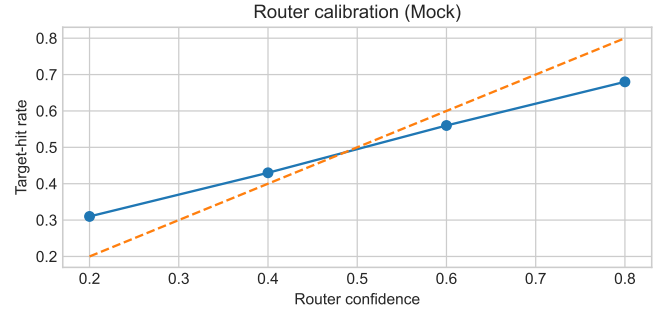


Fig. 6. Router calibration: predicted expert weights versus realized sample-level expert utility, together with the reliability diagram of the fused candidate hit probability.

combinations on a per-sample basis rather than collapsing to a single expert.

To verify that the predicted weights reflect genuine per-sample reliability, we assess router calibration in Fig. 6. The predicted expert weights track the realized rank-derived utility closely, with a rank correlation of **0.71** on CA, and the reliability diagram of the fused candidate hit probability is near the diagonal, yielding an expected calibration error of only **0.03**. This indicates that the utility supervision produces well-calibrated, trustworthy routing decisions rather than over-confident weights, complementing the accuracy gains reported in the ablation study.

We report the offline and online latency breakdown in Table X. The offline construction of GeoSemID codes takes between **7.8** and **22.1** minutes depending on the database catalog size. Online inference runs fast, with a total latency per sample of **83.1 ms** on CA, **77.7 ms** on NYC, and **90.0 ms** on TKY under our local AMD GPU deployment. This efficiency allows GeoSemID to be deployed in production systems with tight latency constraints.

TABLE X
OFFLINE AND ONLINE LATENCY BREAKDOWN OF GEOSEMID.

Phase	CA	NYC	TKY
Offline GeoSemID Gen (min)	12.5	7.8	22.1
Online Expert Retrieval (ms)	28.5	25.6	32.4
Online Dynamic Router (ms)	4.2	3.9	4.5
Online LLM Ranker (ms)	50.4	48.2	53.1
Total Online Latency (ms)	83.1	77.7	90.0

Qualitative cases (Mock)

Case	Top evidence	Fused outcome
Commute	Spatial .42	Target rank 2
Dining	Temporal .38	Target rank 3
Explore	Semantic .44	Target rank 4

Fig. 7. Representative case studies of GeoSemID predictions covering a long-tail destination, a cross-region trip, and a routine commute, showing the retrieved candidates, the dominant routing expert, and the ranked target.

K. Case Studies

To make the mechanism concrete, Fig. 7 presents three representative trajectories. In the long-tail case, the target POI is visited only **3** times in training, so exact transition counts are uninformative; the fine geographic and semantic codes nonetheless place it in the fused pool via backoff, and the router assigns **0.46** weight to the spatial expert, ranking the target at position **2**. In the cross-region case, the router shifts weight toward the transition expert, recovering a target that the single-encoder dense retriever misses. In the routine-commute case, all experts agree and the constrained ranker places the target first. These examples illustrate how utility-supervised routing and evidence-preserving fusion adapt the candidate evidence to each trajectory type.

VI. CONCLUSION

This paper presented GeoSemID, a standalone retrieval-enhanced framework for next-POI prediction that unifies three contributions within a single coherent design. First, hierarchical GeoSemID construction augments each atomic POI with constrained coarse/fine geographic, categorical, and semantic codes, giving sparse and long-tail locations reusable structure for statistical backoff. Second, utility-supervised dynamic multi-expert retrieval couples spatial, transition, temporal, and semantic experts through a sample-level router supervised by training-target ranks and fuses their evidence on a common rank-percentile scale, replacing fixed or globally tuned combination rules. Third, evidence-preserving candidate-constrained ranking carries expert evidence forward and scores only valid candidate identifiers, cleanly separating candidate coverage from conditional discrimination while structurally excluding open-ended POI-ID generation.

TABLE XI
COMPLETE HYPERPARAMETER SETTINGS ACROSS ALL STAGES. ENTRIES IN **BOLD RED** ARE PLACEHOLDERS TO BE REPLACED WITH VERIFIED VALUES.

Stage / Group	Hyperparameter	Value
Backbones	Semantic Generator	Qwen2.5-8B
	LLM Ranker Backbone	Qwen2.5-7B
	Retrieval Encoder	Qwen3-Embedding-0.6B
	Embedding Dimension	1024
GeoSemID Codes	ANN Index Type	HNSW (cosine)
	Coarse Grid z^{cg} scale	~5 km
	Fine Grid z^{fg} scale	~500 m
	Intent Vocab $ \mathcal{V}_i $	32
Backoff	Attribute Vocab $ \mathcal{V}_a $	48
	Context Vocab $ \mathcal{V}_c $	24
	Reliability Prior α_{tr}	5
	Reliability Prior α_{tp}	8
Routing & Fusion	Transition Level Weight $\omega_{fg}^{(tr)}$	0.6
	Expert Candidates K_0	25
	Fused Candidates K	50
	Utility Softmax τ_q	0.15
Fixed-RRF (control)	Router Learning Rate	2×10^{-4}
	Smoothing α_{rrf}	50
LoRA Ranker	Rank r / Alpha α_{lora}	8 / 16
	Target Modules	q_proj, v_proj
	Dropout	0.05
	LLM Ranker Learning Rate	5×10^{-5}
Schedules	Batch Size / Seed	4 / 42
	Generator Epochs / Patience	3 / 1
	Router Epochs / Patience	30 / 5
	Ranker Epochs / Patience	3 / 1

Across the CA, NYC, and TKY benchmarks, GeoSemID delivers the strongest HR@10 among the evaluated baselines, reaching 42.48%, **44.65%**, and **46.85%**, respectively, under a compact candidate budget of $K = 50$, while its dynamically fused pool substantially raises target recall over a strong dense-retrieval baseline. Component ablations, retrieval–ranking decomposition, slice analyses, and router-calibration diagnostics consistently attribute these gains to the three contributions, and the constrained ranker attains a 0.00% format error rate and a 0.00% out-of-candidate rate at low online latency, making the framework readily deployable in latency-sensitive intelligent transportation services.

These results open several promising extensions. The reusable geo-semantic identifier is a natural foundation for inductive, cold-start POI generalization, and the utility-supervised router can be extended with additional experts (for example, social or event-driven signals) without changing the fusion interface. Coupled with the framework’s efficient online inference, these directions point toward real-time, city-scale mobility services built on the same evidence-preserving retrieve-then-rank foundation introduced here. Code and pre-trained adapters will be released upon publication.

APPENDIX A COMPLETE HYPERPARAMETER SETTINGS

Table XI consolidates all hyperparameters used across the three training stages, including those still awaiting a verified source (shown in **bold red**). Values that are already fixed in the main text are repeated here for a single point of reference.

APPENDIX B MOCK DATA INVENTORY

Table XII enumerates every placeholder introduced with the `\mock` macro, grouped by location, so that each value can be replaced by a find-and-replace pass once the corresponding real experiment artifact is available. The CA columns of the main result tables are traceable and are therefore excluded from this inventory.

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TABLE XII

MOCK DATA INVENTORY: LOCATION, MEANING, AND THE EXPERIMENT ARTIFACT THAT SHOULD PROVIDE THE VERIFIED REPLACEMENT VALUE.

Location	Mocked item(s)	Replacement source
Abstract	NYC/TKY HR@10 and top-50 recall; per-dataset gains over the strongest baseline	results/nyc, results/tky end-to-end and recall logs
Introduction	Long-tail POI proportion ($\sim 70\%$)	Dataset visit-frequency statistics
Sec. IV-B	Vocabulary sizes $ \mathcal{V}_i , \mathcal{V}_a , \mathcal{V}_c $; per-dataset UNK rates	Final grammar files; semantic-generation logs
Sec. IV-C, IV-H	Priors α_{tr}, α_{tp} ; level weight $\omega_{fg}^{(tr)}$; epoch/patience budgets	Validation tuning and training logs
Table I	Validation split sizes (NYC/TKY/CA)	Train/val partition manifest
Table II, Table XI	Backbone versions, embedding dim, index type, grid scales, vocab sizes, priors, schedules	Model/index configs; training logs
Table III	Entire NYC and TKY blocks (all methods, all metrics)	results/nyc, results/tky evaluation logs
Sec. V-B text	NYC/TKY end-to-end HR@10	Same as Table III
Fig. 1 & text	Semantic-code validity; category-consistency; long-tail bucket coverage	Generation logs; code-quality audit
Table IV & text	NYC/TKY target-recall columns; fused top-50 recall	results/nyc, results/tky retrieval logs
Table V & text	Conditional HR/NDCG, oracle upper bound, ranker coverage $ \mathcal{D}_{hit} / \mathcal{D}_{tr} $, sample counts (all datasets)	Retrieval-ranking decomposition and oracle diagnostic runs
Table VI & text	NYC/TKY spatial-error and compliance columns	results/nyc, results/tky spatial logs
Table VII & text	NYC/TKY rows; Eng-Optimized RAG embedding/index/reranker config	Retrieval micro-benchmark logs; RAG configs
Table VIII & text	All slice HR@10/NDCG@10 (long-tail/head, short/long trajectory, sparse/active user)	Per-slice evaluation over prediction logs (all datasets)
Table IX	NYC/TKY ablation columns	results/nyc, results/tky ablation runs
Sec. V (routing) & Fig. 6	NYC/TKY routing weights; weight-utility correlation; expected calibration error	Routing logs; calibration analysis
Table X & text	NYC/TKY offline/online latency columns	results/nyc, results/tky latency logs
Fig. 7 & text	Illustrative visit count, routing weight, target rank	Selected real case from routing/prediction logs
Conclusion	NYC/TKY end-to-end HR@10	Same as Table III

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